A Practical Study of Phishing Detection for Safer Web Use

Student Research Proposal • Billal Hossain • Cybersecurity • Network Security • Secure Software

Proposed by	Billal Hossain
Project type	Student-led research / prototype study
Estimated duration	8–10 weeks

Project Overview

This project aims to study how basic machine-learning and security features can be used to distinguish potentially malicious URLs from legitimate ones. The idea came from a simple question: can a lightweight, understandable model provide useful warning signals without requiring a complex security platform?

Student perspective
The project is intentionally scoped so that a small student team can learn, experiment, and produce a useful research artifact without assuming access to a large laboratory or proprietary dataset.

Research Problem

Phishing links can look convincing and may use familiar words, unusual subdomains, URL shortening, or misleading paths. Many detection approaches are difficult for beginners to understand or reproduce. A controlled academic study can compare simple URL-based features and models while clearly documenting the limitations of automated detection.

Research Objectives

- Study common URL characteristics associated with phishing and legitimate websites.
- Prepare a labeled dataset from reputable public sources and document its limitations.
- Engineer interpretable features such as URL length, special-character counts, domain-related attributes, and path patterns.
- Compare simple classifiers such as logistic regression, decision tree, and random forest.
- Investigate false positives and false negatives and identify situations where automated classification can fail.

Research Questions

RQ1. Which URL features are most useful for phishing classification?

RQ2. How does an interpretable baseline compare with an ensemble model?

RQ3. Which phishing URLs produce the most false negatives?

RQ4. Can the system explain its warning signals to a non-expert user?

Proposed Methodology

Step 1. Review phishing indicators and define a feature set that can be extracted without visiting suspicious websites.

Step 2. Clean and balance the dataset where appropriate while keeping the original distribution documented.

Step 3. Train baseline and ensemble classifiers using a reproducible pipeline.

Step 4. Evaluate precision, recall, F1-score, and confusion matrices, with special attention to false negatives.

Step 5. Build a small demonstration interface that explains why a URL was classified as suspicious based on selected features.

Data / Materials / Experimental Resources

The exact resources will be finalized with the collaborator after a feasibility discussion. The preferred approach is to start with public or self-generated material, keep the experiment small enough to reproduce locally, and record the source, preprocessing decisions, assumptions, and version of each important component.

Initial resource plan: Public labeled URL dataset, feature-extraction script, ML training/evaluation notebook.

Evaluation Plan

Evaluation will be based on measurable criteria rather than only a successful demo. Depending on the project, this may include classification metrics, detection quality, latency, error analysis, reproducibility, or qualitative review. Baselines will be established before more complex techniques are introduced so that the contribution of each change can be discussed clearly.

Reproducibility and Documentation

The team will keep a simple experiment log containing dataset versions, preprocessing choices, model or system settings, evaluation results, and important observations. Code will be organized so that the main experiment can be rerun without relying on undocumented manual steps. Charts and tables will be generated from saved results where possible.

Expected Challenges

- Limited time and computing resources may require a smaller experimental scope.
- Dataset quality may affect conclusions more than model complexity.
- A strong-looking metric may hide weaknesses that appear during error analysis.
- The research question may need refinement after an initial feasibility experiment.

Planned Mitigation

The project will begin with a small baseline and a feasibility check. If data or hardware limitations make the original scope unrealistic, the team will narrow the research question rather than artificially inflating results. Each limitation will be reported as part of the final research discussion.

Expected Outcomes

- A documented phishing-URL dataset preparation pipeline.
- Comparison of lightweight ML models for URL classification.
- An interpretable prototype that presents detection signals.
- Security-focused error analysis and recommendations for future work.

Potential Research Collaboration

I would like to work with someone interested in cybersecurity, networking, or applied ML. A collaborator with security research experience could help improve the feature design and make the evaluation more meaningful.

Technical Stack

Python, pandas, scikit-learn, Linux, Git; optional Flask or Node.js for the demo interface.

Scope, Ethics, and Responsible Research

Suspicious URLs will be analyzed as data rather than visited unnecessarily. The project will use safe datasets and controlled experiments and will not be used to conduct phishing, credential theft, or unauthorized security testing.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Setup	Dataset/system preparation and baseline implementation
3. Experiments	Development, comparison, tuning, and documentation
4. Evaluation	Metrics, error analysis, limitations, reproducibility
5. Reporting	Results, discussion, documentation, and presentation

What I Am Looking for in a Collaborator

I am looking for someone who is genuinely interested in the research question and willing to contribute to discussion, implementation, experiment design, data analysis, or writing. I would prefer to divide work according to each person's strengths and keep decisions, code, data, and results documented.

Possible Next Step

If a researcher is interested, I would like to first discuss the research question, available resources, and a realistic scope. After that discussion, the proposal can be refined into a clearer study plan with specific responsibilities and milestones.

Note: This document describes a proposed student project. It does not claim completed research, published results, or guaranteed performance. The scope is open to refinement through collaboration.